

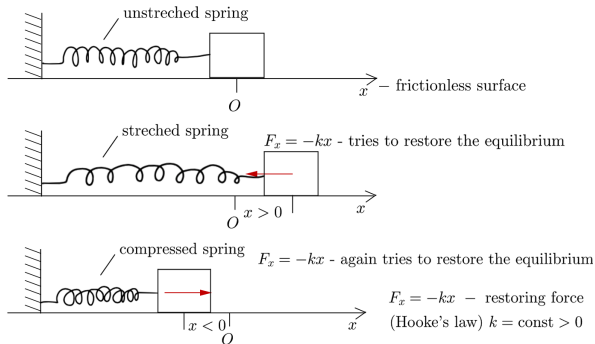
Chapter 6 – Harmonic Oscillator and Mechanical Resonance

UM-SJTU Joint Institute
Physics I (Summer 2024)
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Motivation

Example I: Horizontal Mass-Spring System

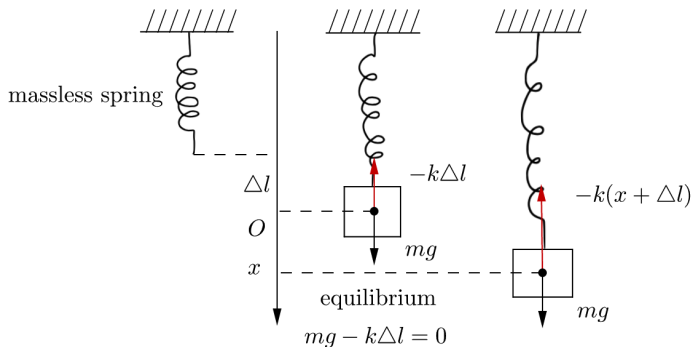


Equation of motion (net force = elastic force)

$$ma_x = F_x \quad \Rightarrow$$

$$\ddot{x} + \frac{k}{m}x = 0$$

Example II. Vertical Mass-Spring System



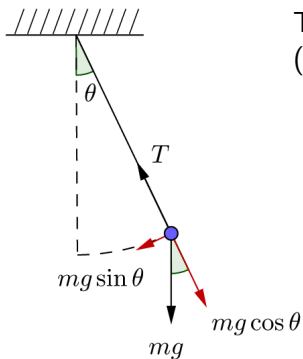
Equation of motion (net force = weight + elastic force)

$$ma_x = mg - k(x + \Delta l) \quad \Rightarrow \quad ma_x = \underbrace{mg - k\Delta l}_{=0} - kx$$

$$ma_x = -kx \quad \Rightarrow$$

$$\ddot{x} + \frac{k}{m}x = 0$$

Example III. Simple Pendulum



Tangential component on the net force
(non-zero contribution only due to weight)

$$F_{\theta} = -mg \sin \theta \approx -mg\theta \quad \uparrow \text{small angles only}$$

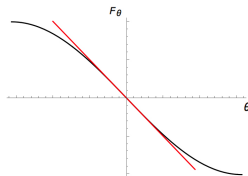
Because

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \dots$$

$\hookrightarrow$ keep the 1st term

When is this approximation valid?

E.g., for $\theta = 0.1$ (6°), $\sin \theta = 0.0998$



Example III. Simple Pendulum

Motion along the arc (tangential components)

$$ma_\theta = F_\theta \quad \overset{a_\theta = l\ddot{\theta}}{\longleftrightarrow} \quad ml\ddot{\theta} \approx -mg\theta$$

$$\ddot{\theta} + \frac{g}{l}\theta = 0$$

Observation:

In all three cases, the equation of motion is of the same form

$$\ddot{x} + \omega_0^2 x = 0$$

[with $\omega_0^2 = k/m$ for the mass-spring systems
and $\omega_0^2 = g/l$ for the simple pendulum]

Any particle (system) with the equation of motion of the above form is called a **simple harmonic oscillator** (SHO).

Simple Harmonic Oscillator

Simple Harmonic Oscillator

Simple Harmonic Oscillator – only the restoring force acts.

$$\ddot{x} + \omega_0^2 x = 0 \quad \implies \quad \boxed{x(t) = ?}$$

How to solve? Guess a solution and check...

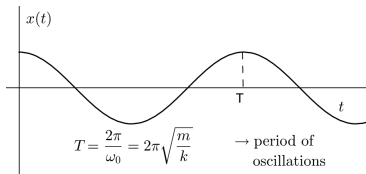
Our guess

$$x(t) = \cos(\omega_0 t)$$

Check

$$\dot{x} = -\omega_0 \sin(\omega_0 t) \quad \text{and} \quad \ddot{x} = -\omega_0^2 \cos(\omega_0 t) = -\omega_0^2 x$$

Periodic behavior
(oscillations)



Have we found the most general form of the solution?

①

$$x(t) = A \cos \omega_0 t$$

$$\dot{x}(t) = -\omega_0 A \sin \omega_0 t$$

$$\ddot{x}(t) = -\omega_0^2 A \cos \omega_0 t = -\omega_0^2 x(t) \quad \rightarrow \text{also solves the equation}$$

②

$$x(t) = A \cos(\omega_0 t + \phi)$$

$$\dot{x}(t) = -\omega_0 A \sin(\omega_0 t + \phi)$$

$$\ddot{x}(t) = -\omega_0^2 A \cos(\omega_0 t + \phi) = -\omega_0^2 x(t) \quad \rightarrow \text{solves it, too!}$$

The most general solution

$x(t) = A \cos(\omega_0 t + \phi)$ amplitude initial phase

Equivalently (see Problem Set 4) the most general solution can be written as

$$x(t) = B \cos \omega_0 t + C \sin \omega_0 t$$

Note: We have two constants (now, B and C) again.

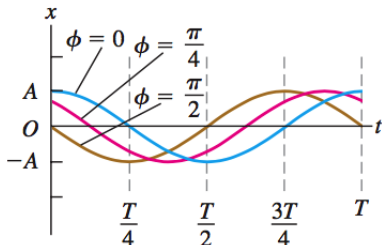
The constants A and ϕ (or B and C) are found by applying the initial conditions:

$$\begin{cases} x(0) = x_0 \\ v_x(0) = v_{0x} \end{cases}$$

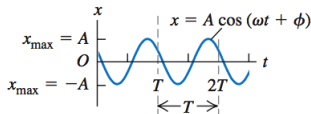
then the problem has a unique solution.

Position, Velocity, and Acceleration in SHM

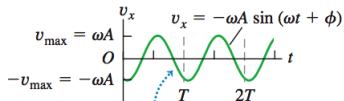
These three curves show SHM with the same period T and amplitude A but with different phase angles ϕ .



(a) Displacement x as a function of time t

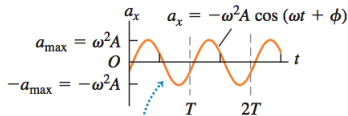


(b) Velocity v_x as a function of time t



The v_x - t graph is shifted by $\frac{1}{4}$ cycle from the x - t graph.

(c) Acceleration a_x as a function of time t



The a_x - t graph is shifted by $\frac{1}{4}$ cycle from the v_x - t graph and by $\frac{1}{2}$ cycle from the x - t graph.

Uniform Circular Motion and Simple Harmonic Motion

Uniform Circular Motion and Simple Harmonic Motion

$$\frac{d\varphi}{dt} = \omega_0 = \frac{v}{R} = \text{const}$$

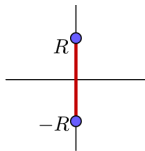
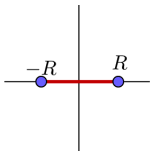
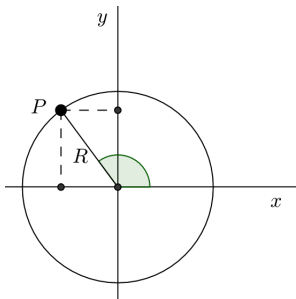
$$\Rightarrow \varphi = \omega_0 t \quad [\text{assume } \varphi(0) = 0]$$

$$x = R \cos \overbrace{\omega_0 t}^{\varphi}, \quad y = R \sin \overbrace{\omega_0 t}^{\varphi}$$

Differentiate twice w.r.t. time

$$\begin{cases} a_x = -R\omega^2 \cos \omega_0 t = -\omega_0^2 x \\ a_y = -\omega_0^2 y \end{cases}$$

Conclusion: The projection of P onto the x axis (or the y axis) moves as if it was in a simple harmonic motion.



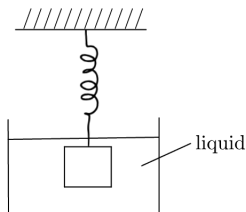
Damped Oscillations

Damped Oscillations

Oscillating object placed in a liquid – add a linear drag to the model ($b = \text{const} > 0$)

$$m \frac{d^2 x}{dt^2} = -b \frac{dx}{dt} - kx$$

$$\ddot{x} + \frac{b}{m} \dot{x} + \omega_0^2 x = 0$$



How to solve? (*not easy to guess directly...*)

Try $x(t) = e^{\lambda t}$ (will need to find λ). Then

$$\dot{x} = \lambda e^{\lambda t} = \lambda x$$

$$\ddot{x} = \lambda^2 e^{\lambda t} = \lambda^2 x$$

Plugging back into the equation of motion

$$\lambda^2 x + \frac{b}{m} \lambda x + \omega_0^2 x = 0 \quad \implies \quad \lambda^2 + \frac{b}{m} \lambda + \omega_0^2 = 0$$

Observation: A differential equation turned into an algebraic (quadratic) one. Easy to solve!

Solution (roots) of the algebraic equation depends on the sign of

$$\Delta = \left(\frac{b}{m}\right)^2 - 4\omega_0^2$$

- ❶ $\Delta < 0$; complex roots $\lambda_{1,2} = -\frac{b}{2m} \pm i\sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}$
- ❷ $\Delta > 0$; real & different roots $\lambda_{1,2} = -\frac{b}{2m} \pm \sqrt{\left(\frac{b}{2m}\right)^2 - \omega_0^2}$
- ❸ $\Delta = 0$; real root (repeated) $\lambda = -\frac{b}{2m}$

Need to analyze these three cases.

Damped Oscillations: Case I (Underdamped Regime)

$$\Delta < 0 \quad \Rightarrow \quad \left(\frac{b}{m}\right)^2 - 4\omega_0^2 < 0 \quad \Rightarrow \quad \boxed{\left(\frac{b}{m}\right)^2 < 4\omega_0^2}$$

It is the case of weak damping (**underdamped regime**). Then, the general solution

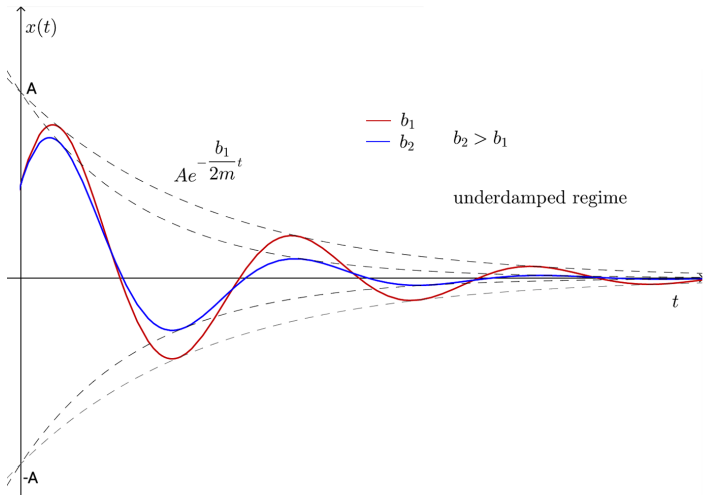
$$\begin{aligned} x(t) &= C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} = \\ &= C_1 e^{-\frac{b}{2m}t} e^{i\sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}t} + C_2 e^{-\frac{b}{2m}t} e^{-i\sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}t} \end{aligned}$$

But $x(t)$ is a physical quantity (displacement from the equilibrium position); must be real, so

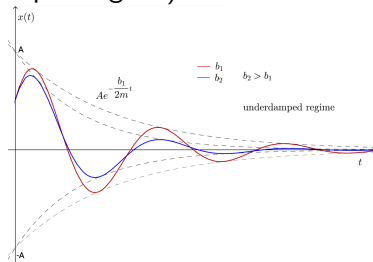
$$C_1 = \frac{1}{2}Ae^{i\phi_0} = C_2^* \quad \Rightarrow \quad C_2 = \frac{1}{2}Ae^{-i\phi_0}$$

Hence (recall the formula: $e^{iu} = \cos u + i \sin u$)...

$$x(t) = Ae^{-\frac{b}{2m}t} \cos \left(\sqrt{\omega_0^2 - \left(\frac{b}{2m} \right)^2} t + \phi_0 \right)$$



Effects of weak damping (underdamped regime)



- * Motion still periodic, but the amplitude of oscillations decreases exponentially with time.

- * The angular frequency of oscillations

$\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2} < \omega_0$, i.e. it is smaller than that for the undamped case (the natural angular frequency ω_0).

Consequently, the period $T = 2\pi/\omega$ increases.

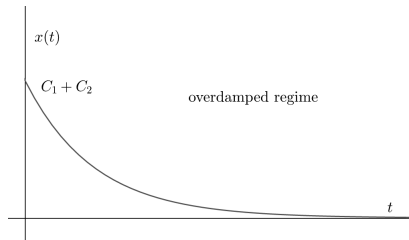
Case II. Overdamped Regime

Now, $\Delta > 0$ so that $\left(\frac{b}{m}\right)^2 > 4\omega_0^2$. It is the case of strong damping (**overdamped regime**) and the general solution

$$\begin{aligned} x(t) &= C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} = \\ &= \boxed{C_1 e^{-\left(\frac{b}{2m} - \sqrt{\left(\frac{b}{2m}\right)^2 - \omega_0^2}\right)t} + C_2 e^{-\left(\frac{b}{2m} + \sqrt{\left(\frac{b}{2m}\right)^2 - \omega_0^2}\right)t}} \end{aligned}$$

Effects of strong damping
(overdamped regime)

- * No periodic behavior.
- * Strong damping results in aperiodic motion: the particle returns aperiodically to the equilibrium position.



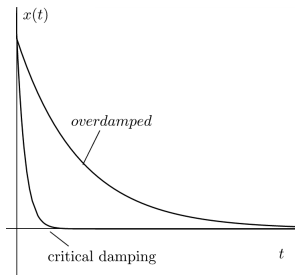
Case III. Critical Damping

Finally, is $\Delta = 0$ so that $\left(\frac{b}{m}\right)^2 = 4\omega_0^2$. In this case damping is called **critical** and the general solution¹

$$x(t) = D_1 e^{-\frac{b}{2m}t} + D_2 t e^{-\frac{b}{2m}t}$$

Effects of critical damping

- * No periodic behavior.
- * The system may pass through the equilibrium position at most once (see Problem Set).



^{1**} The second term includes the factor t in order to make it linearly independent from $e^{-\frac{b}{2m}t}$.

Forced (or Driven) Oscillations. Mechanical Resonance

Forced (or Driven) Oscillations

Now: restoring force + linear drag + **driving force** F_{dr}

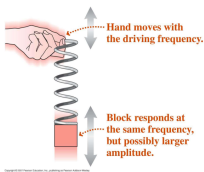
Simplest case to analyze (ω_{dr} – driving frequency):

$$F_{\text{dr}} = F_0 \cos \omega_{\text{dr}} t \quad (\text{sinusoidal time-dependence})$$

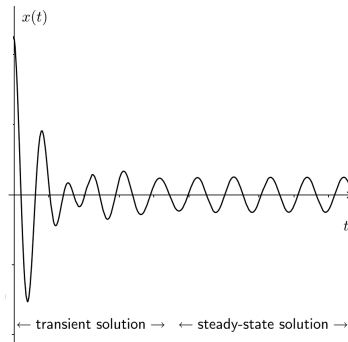
Equation of motion

$$ma_x = F_x = -kx - bv_x + \boxed{F_0 \cos \omega_{\text{dr}} t}$$

$$\ddot{x} + \frac{b}{m} \dot{x} + \underbrace{\frac{k}{m}}_{\omega_0^2} x = \frac{F_0}{m} \cos \omega_{\text{dr}} t$$



Observation: After some time the oscillations stabilize and the particle oscillates with the angular frequency of the driving force (there may be a shift in phase between the drive and the response though).



In general, the solution to the equation of motion in this case is of the form

$$\begin{aligned}
 x(t) = & \boxed{\text{solution to the equation of motion for damped oscillator}} \\
 & \text{(vanishes as } t \rightarrow \infty) \\
 & + \boxed{\text{periodic steady-state oscillations with angular frequency } \omega_{\text{dr}}} \\
 & \quad \quad \quad x_s(t)
 \end{aligned}$$

Steady-State Solution

The steady-state solution

$$x_s(t) = A \cos(\omega_{\text{dr}} t + \phi)$$

ϕ is assumed to be negative, so that it has the interpretation of a *phase-lag*.

Detailed calculations (omitted here²) show that

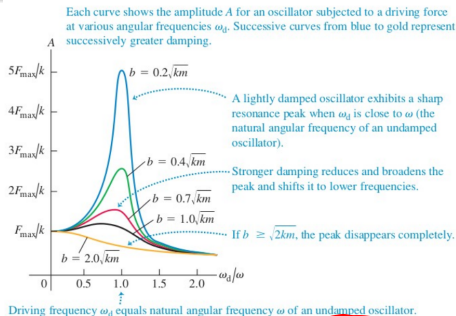
$$A(\omega_{\text{dr}}) = \frac{F_0}{m \sqrt{(\omega_0^2 - \omega_{\text{dr}}^2)^2 + \left(\frac{b\omega_{\text{dr}}}{m}\right)^2}}$$
$$\tan \phi = \frac{b\omega_{\text{dr}}}{m(\omega_{\text{dr}}^2 - \omega_0^2)}$$

Observations. (1) A is a function of ω_{dr} . (2) In general, $\phi \neq 0$, hence F_{dr} and x_s **are not** in phase.

²You are welcome to take Honors Physics next time!

Discussion: Amplitude. Mechanical Resonance

$$A = \frac{F_0}{m\sqrt{(\omega_0^2 - \omega_{dr}^2)^2 + \left(\frac{b\omega_{dr}}{m}\right)^2}}$$

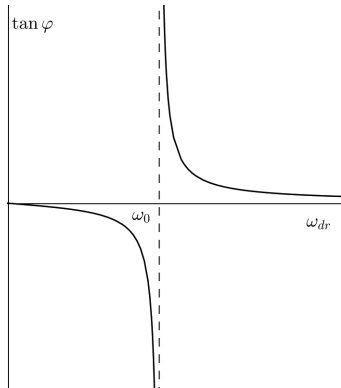
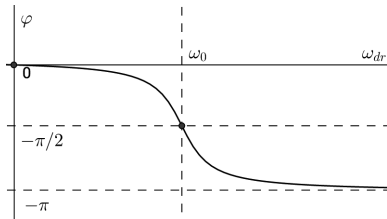


Features

- A peak in the curve $A = A(\omega_{dr})$ at the **resonance frequency**, i.e. for $\omega_{dr} = \omega_{res} = \sqrt{\omega_0^2 - b^2/2m^2}$ [see Problem Set]. The sharp increase in the amplitude of oscillations when $\omega_{dr} \rightarrow \omega_{res}$ is called the **mechanical resonance**.
- Increasing damping shifts the resonance frequency downwards.
- For $\omega_{dr} \rightarrow 0$ (i.e., $T_{dr} \rightarrow \infty$; constant force), then $A \rightarrow \frac{F_0}{m\omega_0^2} = \frac{F_0}{k}$.

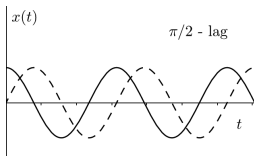
Discussion: Phase-Shift

$$\tan \phi = \frac{b\omega_{dr}}{m(\omega_{dr}^2 - \omega_0^2)}$$



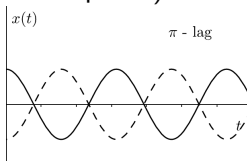
Phase-Shift

- If $\omega_{\text{dr}} \rightarrow \omega_0$ (close to resonance³), then $\phi \rightarrow -\pi/2$.



The response ($x_s(t)$) lags the drive ($F_{\text{dr}}(t)$) by $1/4$ of the cycle.

- If $\omega_{\text{dr}} \rightarrow \infty$ (high frequencies), then $\phi \rightarrow -\pi$ the response lags the drive by $1/2$ of the cycle (displacement and drive are in antiphase)



The response ($x_s(t)$) lags the drive ($F_{\text{dr}}(t)$) by $1/2$ of the cycle.

³Small damping is assumed here, so that $\omega_{\text{res}} \approx \omega_0$.

Mechanical Resonance in Practice. Demonstrations

[videos]